

PLR-1 question statement

you have a main laptop (your laptop). And a friend's laptop (lab PC if needed). Your laptop has wifi, friend's laptop has no wifi. connect both laptops using ethernet cable (will be provided)

you are basically setting up a small LAN with **your laptop** as gateway device . Except that there is no setup to access internet in the LAN, just for now communication between devices in the LAN only

I will give a particular subnet of IP addresses, you will do manual config and assign IPV4 addresses to all the devices in the LAN (most probably just your and friend laptop , could be more devices too)

GOAL : now make sure all devices can ping eachother then run SSH server in your main laptop, from the other connected laptop we'll ssh into your laptop . And FTP server also, we'll access file system of your laptop from your friends laptop using FTP and download files

What you need to show :

1. I will give a particular subnet of ipv4 addresses, you need to use that only. Show on paper what ip addresses, subnet mask you'll use, if ok then you'll connect laptops and assign those ip addresses by manual config in ethernet settings

2. install any ssh and ftp servers already before coming and make sure they're running. From your friend's laptop you'll ssh into your laptop and run a few commands like ls,cd, create files, copy paste,... similarly for FTP, from your friend's laptop you'll FTP login into your laptop, navigate to any particular folder and download a file

commands to do these in linux. (You do your own in windows, research yourself)

to install a ssh server, ftp server : sudo apt install ssh vsftpd

by default they're running when installed, but if necessary manually start the services

sudo systemctl start ssh

sudo systemctl start vsftpd

from friend's system to ssh, ftp login

commands : (for ssh, ftp clients commands are same in any os)

1. ssh username@(ipv4 address). Later enter password when asked

2. ftp (ipv4 address) . Later enter your laptop's username and password

PLR-2 question statement

continuation of above LAN, in your main laptop you setup somethings such that, it acts as a NAT and a router. So your other laptops can access internet through your laptop.

Basically implementing like a mobile sharing network through hotspot

for more context, please explore this

use your mobile as a hotspot device, connect multiple devices to that hotspot. Look at your mobile's ip address and then the device's ip addresses. Your mobile is basically providing a NAT, and also acting as a router

GOAL : make your laptop work as a router and a NAT device so that your friend laptop can access internet through your laptop. Your and your friend's laptop, if possible more systems connected together will form a LAN

we already assigned ip addresses from the given subnet to all necessary devices. We just need to allow traffic forwarding from your laptop's ethernet interface to wifi interface

in ubuntu I had to run these commands for that

```
#!/bin/bash
```

```
# Enable Linux to forward packets between Ethernet and Wi-Fi
```

```
sudo sysctl -w net.ipv4.ip_forward=1
```

```
# NAT Windows traffic through the Wi-Fi interface
```

```
sudo iptables -t nat -A POSTROUTING -s 192.168.10.0/30 -o wlo1 -j MASQUERADE
```

```
# Allow Windows -> Ethernet -> Wi-Fi -> Internet
```

```
sudo iptables -A FORWARD -i eno1 -o wlo1 -j ACCEPT
```

```
# Allow Internet replies -> Wi-Fi -> Ethernet -> Windows
```

```
sudo iptables -A FORWARD -i wlo1 -o eno1 \
-m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
```

but for your windows laptops you can use ICS feature and do this by GUI only – **left for you to learn and implement**

What you need to show :

set up your main laptop and show that your friend's laptop can access internet through yours